Ce Zhang

cezhang@cs.unc.edu | ceezh.github.io | twitter.com/cezhhh | Google Scholar

Education

UNC-Chapel Hill — Chapel Hill, NC, US
Ph.D. student in Computer Science — *Aug. 2023 – Present*
- Advisor: Prof. Gedas Bertasius

Brown University — Providence, RI, US
M.S. in Computer Science — *Aug. 2021 – May 2023*
- Advisor: Prof. Chen Sun

Southeast University — Nanjing, Jiangsu, China
B.S. in Computer Science — *Aug. 2016 – Jun. 2020*

Research Interest

I'm broadly interested in Computer Vision and Multimodal AI, particularly **long-video understanding and reasoning**. My research focuses on developing intelligent agents that can efficiently perceive, search, and reason over long videos. Recently, I'm especially interested in **agentic video understanding, long-term multimodal memory, and adaptive information retrieval**, aiming to enable AI systems to effectively understand and reason over long-context multimodal information.

Academic Experience

Research Assistant — Aug. 2023 – Present
UNC-Chapel Hill, Advisor: Prof. Gedas Bertasius — *Chapel Hill, NC*
- Proposed VideoTreeSearch (VTS), a self-correcting agentic framework for grounded long-video QA. VTS represents a long video as an adaptive temporal tree and navigates it via discrete actions (zoom_in, zoom_out, shift, answer), enabling explicit backtracking and recovery. VTS outperforms prior agentic methods across Grounded LVQA benchmarks (CG-Bench, Haystack-LVBench, Haystack-Ego4D) and transfers to general long-video QA (Video-MME, MLVU, LVBench). (**NeurIPS, 2026**) paper link
- Proposed SiLVR, a training-free language-based video reasoning framework that transforms video and speech into language and applies LLMs for video reasoning. SiLVR achieves state-of-the-art performance on multiple video reasoning benchmarks such as Video-MMMU and EgoLife. (**TMLR**) paper link
- Proposed a framework for long-range videoQA by decomposing videoQA into short-term visual captioning and long-range language modeling. Achieved SOTA zero-shot videoQA performance on EgoSchema, NeXT-QA, IntentQA and NeXT-GQA. (**EMNLP 2024**) paper link

Research Assistant — Apr. 2022 – May 2023
Brown University, Advisor: Prof. Chen Sun — *Providence, RI*
- Represent the video with discretized action labels and utilized LLMs for reasoning. Achieved SOTA long-term action anticipation performance on Ego4D, EPIC-Kitchens-55 and EGTEA Gaze+. (**ICLR 2024**) paper link
- Extracted task-specific object-centric representations from pretrained models (e.g. GLIP). Utilized object-centric representations for long-term action anticipation. Achieved competitive results on Ego4D, 50Salads and EGTEA Gaze+. (**WACV 2024**) paper link
- Condensed expert trajectory demonstrations into useful representations for policy learning. Achieved competitive performance on AntMaze, FrankaKitchen and Locomotion. (**NeurIPS 2023**) paper link

Industry Experience

Research Scientist Intern — May 2026 – Aug. 2026
Meta FAIR, Mentor: Ramakanth Pasunuru — *Seattle, WA*
- Worked on parametric memory for long-document understanding. Conducted empirical analyses comparing explicit and parametric memory, and designed a pipeline that integrates both approaches.

Research Scientist Intern — May 2025 – Aug. 2025
Google DeepMind, Mentor: Dahun Kim — *Mountain View, CA*

- Worked on long-form video temporal grounding and question answering. Designed and implemented a training-free, multi-tier coarse-to-fine framework that significantly improves the temporal grounding ability of multimodal large language models (MLLMs), achieving both higher effectiveness and efficiency.

Research Scientist Intern May 2024 – Dec. 2024

Meta FAIR, Mentor: Satwik Kottur *Menlo Park, CA*

- Augmented long-horizon video-based planning ability of large VLMs. Designed multiple auxiliary tasks (e.g., goal prediction, state prediction) and trained the model on all tasks jointly. Leveraged Multi-token Prediction to model the structured action space. Achieved SOTA performance on COIN, CrossTask and Ego4D.

Publication

- *Baiqi Li, Ce Zhang, Yu Fang, Yue Yang, Shangzhe Li, Mingyu Ding, Gedas Bertasius*
  WatchAct: A Benchmark for Behavior-Grounded Robot Manipulation **(arXiv 2026)** paper link
- *Ce Zhang, Ziyang Wang, Yulu Pan, Oluwatumininu Oguntola, Pranav Wagh, Qiyu Wu, Hiromi Wakaki, Mohit Bansal, Gedas Bertasius*
  Searching Videos as Trees: Self-Correcting Agents for Grounded Long Video QA **(NeurIPS 2026)** paper link
- *Ziyang Wang*, Yue Zhang*, Shoubin Yu, Ce Zhang, Zengqi Zhao, Jaehong Yoon, Hyunji Lee, Gedas Bertasius, Mohit Bansal*
  EgoMemReason: A Memory-Driven Reasoning Benchmark for Long-Horizon Egocentric Video Understanding **(COLM 2026)** paper link
- *Baiqi Li, Kangyi Zhao, Ce Zhang, Chancharik Mitra, Jean de Dieu Nyandwi, Gedas Bertasius*
  TimeBlind: A Spatio-Temporal Compositionality Benchmark for Video LLMs **(EMNLP 2026)** paper link
- *Ce Zhang, Yan-Bo Lin, Ziyang Wang, Mohit Bansal, Gedas Bertasius.*
  SiLVR: A Simple Language-based Video Reasoning Framework **(TMLR)** paper link
- *Ce Zhang, Yale Song, Ruta Desai, Michael Louis Iuzzolino, Joseph Tighe, Gedas Bertasius, Satwik Kottur.*
  Enhancing Visual Planning with Auxiliary Tasks and Multi-token Prediction **(WACV 2025)** paper link
- *Yulu Pan, Ce Zhang, Gedas Bertasius.*
  BASKET: A Large-Scale Video Dataset for Fine-Grained Skill Estimation **(CVPR 2025)** paper link
- *Ce Zhang*, Taixi Lu*, Md Mohaiminul Islam, Ziyang Wang, Shoubin Yu, Mohit Bansal, Gedas Bertasius.*
  A Simple LLM Framework for Long-Range Video Question-Answering **(EMNLP 2024)** paper link
- *Qi Zhao*, Shijie Wang*, Ce Zhang, Changcheng Fu, Minh Quan Do, Nakul Agarwal, Kwonjoon Lee, Chen Sun*
  AntGPT: Can Large Language Models Help Long-term Action Anticipation from Videos? **(ICLR 2024)** paper link
- *Ce Zhang*, Changcheng Fu*, Shijie Wang, Nakul Agarwal, Kwonjoon Lee, Chiho Choi, Chen Sun*
  Object-centric Video Representation for Long-term Action Anticipation **(WACV 2024)** paper link
- *Zilai Zeng, Ce Zhang, Shijie Wang, Chen Sun*
  Goal-Conditioned Predictive Coding as an Implicit Planner for Offline Reinforcement Learning **(NeurIPS 2023)** paper link

Services

Conference Reviewer: NeurIPS 2026, CVPR 2026, NeurIPS 2025, ICCV 2025, CVPR 2025, ECCV 2024 (**Outstanding Reviewer Award**), ACL Rolling Review (May 2025, Dec. 2024, June 2024)
Journal Reviewer: TPAMI, TCSVT
Workshop Organizer: LOVEU @ CVPR 2026, T4V @ CVPR 2026, T4V @ CVPR 2025, T4V @ CVPR 2024